Statistical Analysis Report

Descriptive statistics, correlation analysis and hypothesis testing

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Scope and data caveats

This report is one of four companion reports for the Data Science track (Data Quality, Exploratory Data Analysis, Statistical Analysis, Predictive Analytics), all produced by the executed notebook HealthConnect_DataScience_Notebook.ipynb. Five tables named in the brief were not supplied (Health Facilities master, Laboratory Tests, Prescriptions, Admissions, Referrals); only 5,000 of 60,000 bills, diagnoses and payments link to an appointment; and the data behave like simulated data. Analyses that depend on missing tables use labelled proxies or are marked not computable.

Statistical analysis

Descriptive statistics

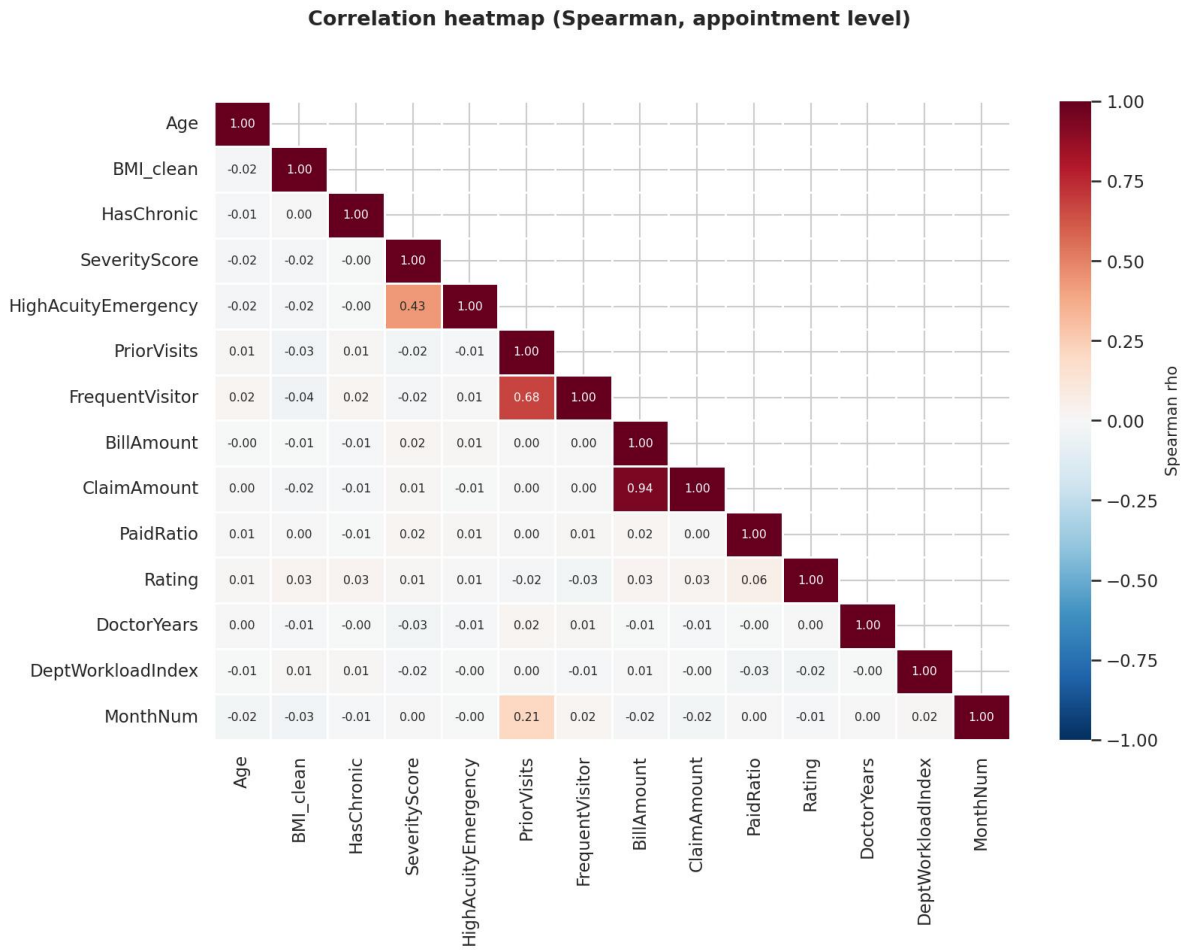
Variable	n	Mean	Median	Mode	SD	Min	Q1	Q3	Max	Skew
Patient age (years)	20,000	45.03	45.0	90	26.26	0.0	22.0	68.0	90.0	0.01
BMI (adults 18+)	16,030	30.04	29.2	32	10.42	11.1	21.7	37.0	61.2	0.45
Bill amount (RWF)	60,000	155156.70	155,247	36,833	83,840	10,002	82,275	227,908	299,998	-0.00
Amount paid (RWF)	60,000	139824.24	129,822	129,528	83,370	5,002	68,694	210,240	299,998	0.26
Outstanding per bill (RWF)	60,000	15332.46	0.0	0	36,038	0.0	0.0	0.0	149,991	2.33
Insurance claim amount (RWF)	42,036	116135.30	111,422	42,625	67,812	5,173	59,154	165,175	298,533	0.33
Satisfaction rating (1-5)	21,064	2.99	3.0	2	1.41	1.0	2.0	4.0	5.0	0.01
Staff years of experience	1,250	12.93	13.0	3	7.17	1.0	6.0	19.0	25.0	0.02
Appointments per doctor	350	14.29	14.0	15	3.68	5.0	12.0	16.0	28.0	0.37
Prior visits (per appointment)	5,000	0.12	0.0	0	0.36	0.0	0.0	0.0	3.0	3.07
Severity score (1-3)	5,000	1.99	2.0	2	0.81	1.0	1.0	3.0	3.0	0.02

Variance, percentiles (5th, 10th, 90th, 95th), IQR and kurtosis are in the notebook (section 6.1). Age, bill amount and rating have skewness near 0 and kurtosis near -1.2, the signature of a uniform distribution, so rank-based tests are used as the primary tests.

Correlation analysis

Relationship	Method	n	Statistic	p
Age vs admission proxy (high-acuity emergency)	Spearman	5,000	-0.017	0.232
Age vs billing amount	Spearman	5,000	-0.003	0.842
Diagnosis vs severity (substitute for 'diagnosis vs length of stay')	Cramer's V	5,000	0.033	0.214
Diagnosis vs billing amount	Kruskal-Wallis (H)	5,000	3.018	0.555
Department workload index vs mean satisfaction (10 department types)	Spearman	10	0.382	0.276
Facility appointments per clinical staff vs mean satisfaction (20 facilities)	Spearman	20	0.028	0.907
Revenue vs patient volume - management summary (12 months)	Spearman	12	0.329	0.297
Billed revenue vs appointments - linked data (12 months)	Spearman	12	0.874	<0.001
Billed revenue vs appointments by facility (20 facilities)	Spearman	20	0.861	<0.001
Insurance plan (15) vs visits per patient (utilisation)	Kruskal-Wallis (H)	20,000	10.931	0.691
Coverage category (approved/pending/self-pay) vs satisfaction	Kruskal-Wallis (H)	1,756	1.474	0.478

Diagnosis vs length of stay cannot be tested (no admissions); diagnosis vs severity and billing are substituted. The two strong correlations (billed revenue vs appointments, monthly and by facility) are structural: each appointment generates one bill of similar size.



Interpretation: Apart from four structural pairs (visits count vs frequent-visitor flag; claim amount vs bill amount, as claims are 50-100% of the bill; severity vs the admission proxy that is built from it; prior visits vs month, which can only grow as the year progresses), no relationship is meaningful: the strongest remaining pair, PaidRatio vs Rating, has |rho| = 0.06. Age, chronic status, severity, billing and satisfaction are effectively independent in this dataset.

Figure: Correlation heatmap (Spearman, appointment level)

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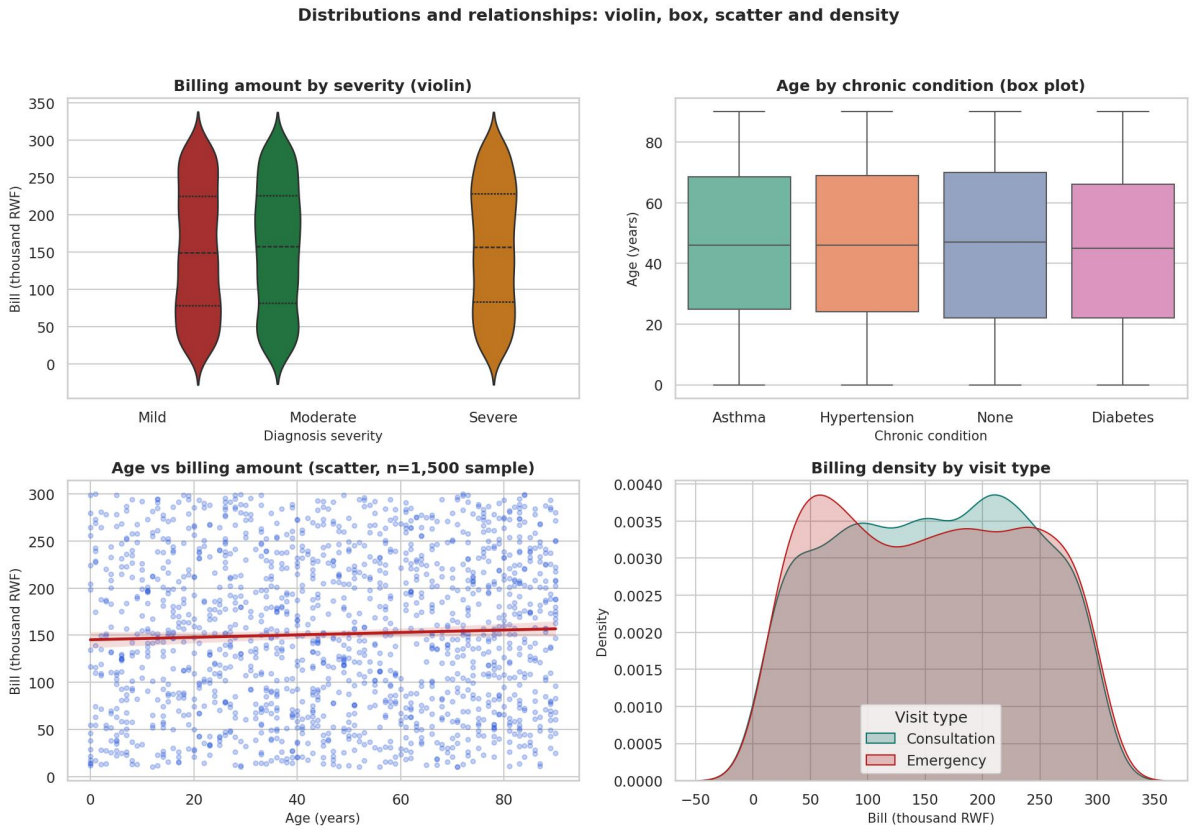
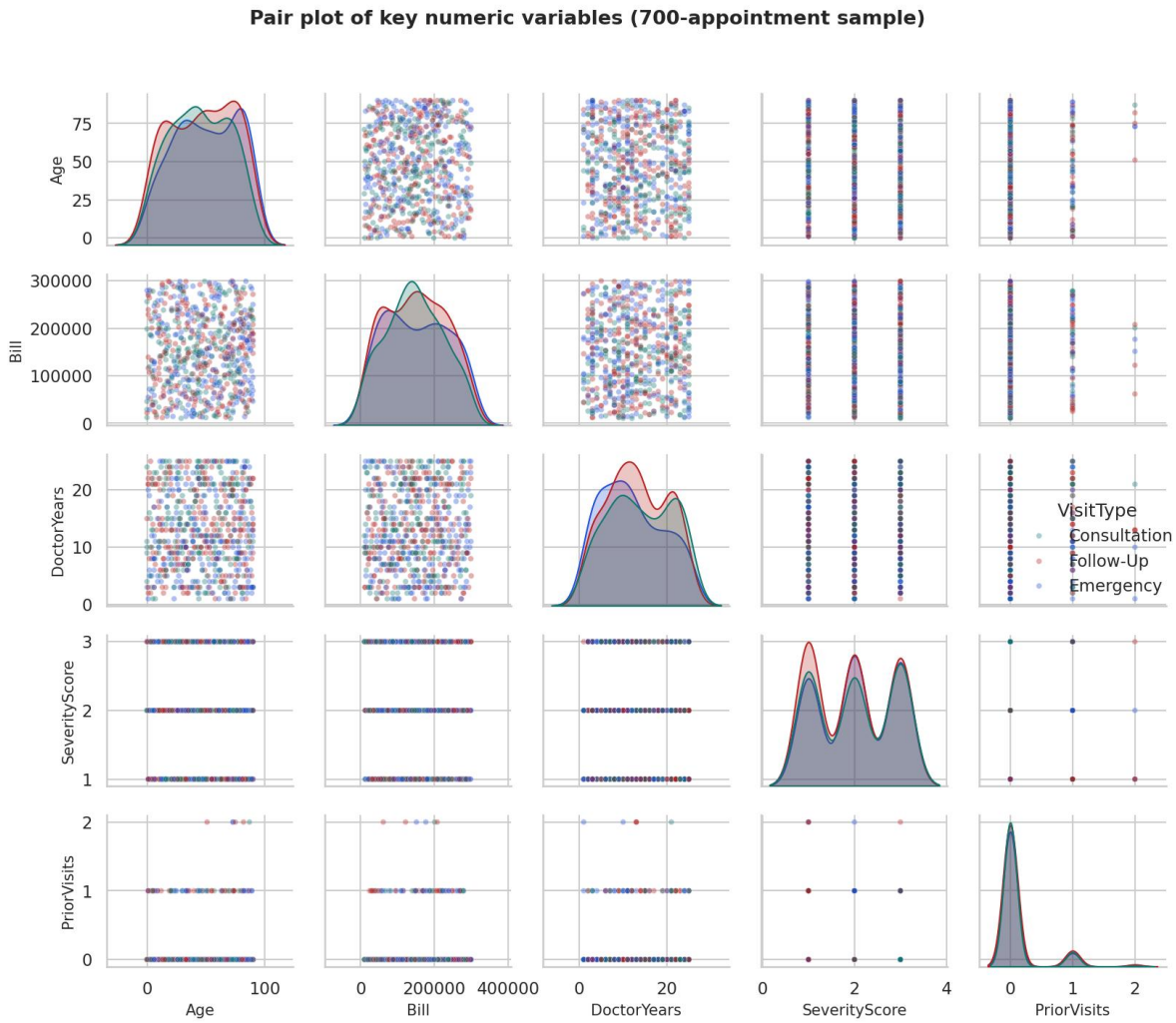


Figure: Distributions and relationships: violin, box, scatter and density

Bills have the same distribution at every severity level and visit type, age has the same distribution for every chronic condition, and the age-billing regression line is flat (Spearman rho=-0.003, p=0.84). In real data, sicker and older patients would cost more; here cost is unrelated to either.



Interpretation: No pair of variables shows a visible pattern or clustering by visit type; points fill each rectangle evenly. Discrete variables (severity score, prior visits) form horizontal and vertical bands only because they take few values.

Figure: Pair plot of key numeric variables (700-appointment sample)

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Hypothesis tests

Significance level 0.05 with Holm correction across all 11 tests. Insured vs uninsured (Test 1) is adapted: insured means the bill has an insurance claim.

Test, hypotheses and effect	Method	p	Holm p	Decision
1. Billing: insurance-backed vs self-pay bills H0: Bill amounts have the same distribution for insured and self-pay bills H1: Distributions differ Effect: mean diff -1,065 RWF (95% CI -2,531 to 401); AUC=0.496; Welch p=0.154	Mann-Whitney U (Welch t as check)	0.155	1.000	Fail to reject H0

Test, hypotheses and effect	Method	p	Holm p	Decision
2. Satisfaction differs across the 20 facilities H0: Median rating is equal in all facilities H1: At least one facility differs Effect: epsilon^2=0.0000; ANOVA p=0.785	Kruskal-Wallis (ANOVA as check)	0.795	1.000	Fail to reject H0
3. Admission proxy rate differs between age groups H0: Proportion of high-acuity emergency visits is equal across age groups H1: Proportions differ Effect: Cramer's V=0.035; rates <18: 10.5%, 18-39: 12.8%, 40-59: 10.8%, 60+: 9.9%	Chi-square test of independence	0.104	0.935	Fail to reject H0
4. Satisfaction differs across department types H0: Median rating is equal in all 10 department types H1: At least one department differs Effect: epsilon^2=0.0000	Kruskal-Wallis	0.878	1.000	Fail to reject H0
5. Chronic-condition prevalence differs by age group H0: Prevalence is the same in all 9 age groups H1: Prevalence differs Effect: Cramer's V=0.018	Chi-square test of independence	0.570	1.000	Fail to reject H0
6. Chronic-condition prevalence differs by sex H0: Prevalence is equal for males and females H1: Prevalence differs Effect: male 74.8% vs female 74.5%	Chi-square test of independence	0.653	1.000	Fail to reject H0
7. Chronic-condition prevalence differs by province H0: Prevalence is equal in all 5 provinces H1: Prevalence differs Effect: Cramer's V=0.022	Chi-square test of independence	0.046	0.502	Fail to reject H0
8. Billing differs by diagnosis severity H0: Bill distribution is the same for Mild, Moderate and Severe H1: At least one differs Effect: epsilon^2=0.0000	Kruskal-Wallis	0.394	1.000	Fail to reject H0
9. Payment completeness differs by Billing.Status label H0: Share of the bill paid is the same for Paid / Partially Paid / Pending bills H1: Status labels differ in share paid Effect: mean paid share: Paid: 90.0%, Partially Paid: 90.1%, Pending: 90.2%	Kruskal-Wallis	0.643	1.000	Fail to reject H0
10. Appointment volume is uneven across facilities H0: Appointments are spread equally over the 20 facilities H1: Volume differs by facility Effect: range 217-278 (expected 250)	Chi-square goodness of fit	0.052	0.521	Fail to reject H0
11. Appointment volume is uneven across department types H0: Appointments are spread equally over the 10 department types H1: Volume differs by department Effect: range 467-544 (expected 500)	Chi-square goodness of fit	0.490	1.000	Fail to reject H0

Reading the results. No test survives correction. Test 7 (chronic prevalence by province) has a raw p of 0.046 but a Holm-adjusted p of 0.50, a good example of why uncorrected results should not be trusted. Test 9 is the most useful: the share of a bill actually paid is the same (about 90%) whether the bill is labelled Paid, Partially Paid or Pending.